Lecture 6: Entropy and The Second Law of Thermodynamics

Lecturer: Amna Mohamed

1. Introduction

The First Law of Thermodynamics establishes that energy is conserved in any thermodynamic process. However, it does not predict or restrict the ***direction*** of a process. For instance, heat naturally and spontaneously flows from a hot object to a cold one, but it never flows spontaneously from a cold object to a hot one, even though such an inverse process would still perfectly conserve energy.

Therefore, the First Law lacks a criterion for spontaneity and directional legality. The **Second Law of Thermodynamics** addresses this limitation by introducing a foundational criterion for the direction of natural processes through a fundamental state property known as **Entropy**.

2. Statements of the Second Law

To fully grasp the scope of the Second Law, we look at the two classical, historically significant statements concerning the boundaries of heat engines and refrigerators:

A. The Kelvin-Planck Statement

It is impossible to construct a heat engine that operates in a thermodynamic cycle and produces no other effect than the extraction of heat from a single thermal reservoir and the performance of an equivalent amount of work. In simpler terms, no heat engine can convert all absorbed heat completely into useful work; therefore, a thermal efficiency of $\eta = 100\%$ is physically impossible.

B. The Clausius Statement

It is impossible to construct a cyclic device that operates with no other effect than the continuous transfer of heat from a cooler body to a hotter body. This means that heat cannot pass spontaneously from a cold reservoir to a hot reservoir without the input of external mechanical work.

3. Thermodynamic Temperature (The Kelvin Scale)

The absolute thermodynamic temperature scale is fundamentally independent of the thermometric properties of any specific substance. It is derived directly from the operating mechanics of a reversible (Carnot) engine working between two thermal reservoirs at absolute temperatures T_H (hot reservoir) and T_C (cold reservoir).

For any reversible engine, the ratio of heat transferred is directly proportional to the ratio of absolute temperatures:

$$Q_H / Q_C = T_H / T_C$$

Where:

- Q_H: Heat absorbed from the hot reservoir.
- Q_C: Heat rejected to the cold reservoir.
- T_H, T_C: Absolute temperatures measured in Kelvin (K).

Example 1 (مثال تطبيقي مباشر)

A reversible Carnot engine operates between a hot reservoir at $600\ K$ and a cold reservoir at $300\ K$. If the engine absorbs $2000\ J$ of heat from the hot reservoir, calculate the heat rejected to the cold reservoir.

Solution:

Given parameters: $T_H = 600\ K$, $T_C = 300\ K$, $Q_H = 2000\ J$

Applying the thermodynamic temperature ratio relationship:

$$Q_H / Q_C = T_H / T_C \Rightarrow 2000 / Q_C = 600 / 300$$

$$2000 / Q_C = 2 \Rightarrow Q_C = 2000 / 2 = 1000\ J$$

Answer: The heat rejected to the cold reservoir is **1000 J**.

4. The Concept of Entropy (S)

Entropy (S) is an intrinsic thermodynamic state function that quantitatively measures the degree of molecular disorder, randomness, or chaotic distribution within a system. From a macroscopic view, Rudolph Clausius defined the exact change in entropy (dS) during an infinitesimal, reversible process as the ratio of the heat exchange to the absolute temperature at which that exchange takes place:

$$dS = dQ_{rev} / T$$

- **SI Unit:** Joules per Kelvin (J/K).
- Entropy is a **state function**; its net change depends entirely on the initial and final equilibrium states, completely independent of the path taken.
- For any completely **reversible cyclic process**, the Clausius expression yields: $\oint dQ/T = 0$.

5. Calculation of Entropy Change (ΔS) in Reversible Processes

To determine the definitive net change in entropy ΔS between two macroscopic states (1 and 2), we integrate the Clausius definition along an ideal reversible path:

$$\Delta S = S_2 - S_1 = \int_{1}^{2} (dQ_{rev} / T)$$

Case A: Isothermal Process (Constant Temperature, $T = ext{constant}$)

Because the absolute temperature remains unchanged throughout the process, it can be mathematically extracted from the integral:

$$\Delta S = (1 / T) \int_{1}^{2} dQ_{rev} = Q_{rev} / T$$

Example 2 (مثال تطبيقي مباشر)

An ideal gas undergoes a reversible isothermal expansion at a constant temperature of ***400 K***. During this process, the gas absorbs ***1200 J*** of heat. Calculate the change in entropy of the gas.

Solution:

Given parameters: $T = 400\ K$, $Q_{rev} = +1200\ J$ (positive value since heat enters the system)

Using the isothermal entropy formula:

$$\Delta S = Q_{rev} / T = 1200 / 400 = 3\ J/K$$

Answer: The change in entropy of the gas is **+3 J/K** (indicating an increase in molecular disorder).

Case B: Isochoric Process (Constant Volume, $V = ext{constant}$)

The heat added reversibly is given by $dQ_{rev} = n\ C_v\ dT$, where n is the number of moles and C_v is the molar heat capacity at constant volume. Integrating yields:

$$\Delta S = n\ C_v\ \ln(T_2 / T_1)$$

Example 3 (مثال تطبيقي مباشر)

A $2\ ext{moles}$ sample of an ideal gas is heated at a constant volume from an initial temperature of $300\ K$ to a final temperature of $600\ K$. Given that the molar heat capacity at constant volume is $C_v = 12.5\ J/mol·K$, find the change in entropy.

Solution:

Given parameters: $n = 2$, $T_1 = 300\ K$, $T_2 = 600\ K$, $C_v = 12.5\ J/mol·K$

Applying the isochoric formula:

$$\Delta S = n\, C_v \ln(T_2 / T_1) = 2 \times 12.5 \times \ln(600 / 300)$$

$$\Delta S = 25 \times \ln(2) = 25 \times 0.693 = 17.33\ J/K$$

Answer: The change in entropy is **+17.33 J/K**.

Case C: Isobaric Process (Constant Pressure, $P = ext{constant}$)

The heat added reversibly is given by $dQ_{rev} = n\,C_p\,dT$, where C_p is the molar heat capacity at constant pressure. Integrating yields:

$$\Delta S = n\,C_p\,\ln(T_2 / T_1)$$

Example 4 (مثال تطبيقي مباشر)

A $1\ ext{mole}$ sample of an ideal gas is cooled at a constant pressure from $400\ K$ to $200\ K$. If the molar heat capacity at constant pressure is $C_p = 20.8\ J/mol{\cdot}K$, calculate the change in entropy.

Solution:

Given parameters: $n = 1$, $T_1 = 400\ K$, $T_2 = 200\ K$, $C_p = 20.8\ J/mol{\cdot}K$

Applying the isobaric formula:

$$\Delta S = n\,C_p\,\ln(T_2 / T_1) = 1 \times 20.8 \times \ln(200 / 400)$$

$$\Delta S = 20.8 \times \ln(0.5) = 20.8 \times (-0.693) = -14.41\ J/K$$

Answer: The change in entropy is **-14.41 J/K** (The negative sign indicates entropy decreased because heat was released by the system into the surroundings).

Case D: Reversible Adiabatic Process (No Heat Exchange, $dQ_{rev} = 0$)

Since there is zero heat transfer into or out of the system boundaries during a reversible adiabatic path:

$$dS = 0 / T = 0 \Rightarrow \Delta S = 0$$

Note for Students: *A process that is both entirely reversible and adiabatic is called an* ***isentropic process*** *(a constant-entropy process).*

Example 5 (مثال تطبيقي مباشر)

An ideal gas expands reversibly and adiabatically inside an insulated thermodynamic cylinder. What is the total change in entropy (***ΔS***) for the gas during this expansion process?

Solution:

By definition, an adiabatic process means that the net heat transfer is zero (***Q_{rev} = 0***). Since the path is explicitly stated as reversible, we use the entropy relation directly:

$$\Delta S = \int (dQ_{rev} / T) = 0$$

Answer: The change in entropy ***ΔS = 0***.

6. Student Evaluation: Multiple Choice Questions (MCQs)

This evaluation section contains curated multiple-choice questions with full explanations to measure understanding of the core lecture themes.

Q1. A system releases 500 J of thermal energy to its surroundings during a reversible isothermal process maintained at a temperature of 250 K. What is the entropy change of the system?

- ☐ +2.0 J/K
- ☑ **-2.0 J/K**
- ☐ +125,000 J/K
- ☐ -125,000 J/K

Explanation: *Because the system releases heat, Q is negative (-500 J). Using ΔS = Q/T yields -500 / 250 = -2.0 J/K.*

Q2. Which of the following thermodynamic statements correctly defines a reversible adiabatic process?

- ☐ It is an isothermal process where ΔS is always greater than zero.
- ☐ It is an isobaric process where ΔS is less than zero.
- ☑ **It is an isentropic process where ΔS is strictly equal to zero.**
- ☐ It is an isochoric process where work done is equal to heat added.

Explanation: *A reversible adiabatic process involves zero heat transfer, making dS = 0. Therefore, entropy remains constant, which is the definition of an isentropic process.*

Q3. According to the Kelvin-Planck statement of the Second Law of Thermodynamics, which of the following is completely impossible?

- ☐ Operating a refrigerator without putting in mechanical work.
- ☑ **Constructing a heat engine that features a thermal efficiency of exactly 100%.**
- ☐ Transferring heat from a cold body to a hot body via an input of work.
- ☐ Allowing heat to flow spontaneously from a hot body to a cold body.

Explanation: *The Kelvin-Planck statement explicitly states that no engine can convert 100% of absorbed heat into net work; some energy must always be rejected to a lower-temperature reservoir.*